Circles

Test 1

Date: _____ Duration: 1 hour 30 minutes Maximum marks: 40

All questions are compulsory. Draw a rough diagram wherever it helps. Show all your steps and give reasons.

Section A (1 mark each)

Questions 1, 2, 3 and 5 are multiple choice. Question 4 is fill in the blank. Question 6 is assertion–reason.

1. In a circle with centre O, chords AB and CD are equal and $\angle OAB = 55^\circ$. What is $\angle COD$? [1]

(a) 55° (b) 70°

(c) 110° (d) 125°

2. A chord of length 20 cm is drawn in a circle of radius 26 cm. What is the distance of the chord from the centre? [1]

(a) 6 cm (b) 16 cm

(c) 24 cm (d) 28 cm

3. Which statement about a circle with centre O is **false**? [1]

(a) The line joining O to the midpoint of a chord (not a diameter) is perpendicular to the chord.

(b) The perpendicular from O to a chord bisects the chord.

(c) The perpendicular bisector of every chord passes through O.

(d) Every line through O bisects every chord of the circle.

4. Fill in the blank: The number of circles that pass through three points lying on one straight line is _____. [1]

5. A circle has radius 10 cm. It has two chords of lengths 12 cm and 18 cm. Which chord is nearer the centre? [1]

(a) The 12 cm chord (b) The 18 cm chord

(c) Both are at the same distance (d) It cannot be decided without more information

6. Assertion–reason. *Choose: (a) Both A and R are true, and R is the correct explanation of A. (b) Both A and R are true, but R is not the correct explanation of A. (c) A is true, but R is false. (d) A is false, but R is true.*

Assertion (A): In a circle with centre O, $OM \perp AB$, where AB is a chord and M lies on AB. If $AM = 5$ cm, then $AB = 10$ cm.

Reason (R): The perpendicular from the centre of a circle to a chord bisects the chord. [1]

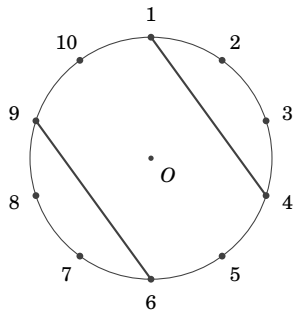
Section B (2 marks each)

7. In a circle with centre O, chord $AB = 10$ cm. P is a point on AB with $AP = 4$ cm. Meena writes: "OP is perpendicular to AB, because the line joining the centre to a point of a chord is perpendicular to the chord."

(a) What is wrong in Meena's reasoning? [1]

(b) Which point Q of AB must be joined to O so that $OQ \perp AB$? How far is Q from A? [1]

8. Divya draws a circle with centre O for a rangoli. She marks 10 dots on the circle at equal distances and numbers them 1 to 10 in order. She joins dot 1 to dot 4, and dot 6 to dot 9 with straight lines.



(a) Find the angle that the line from dot 1 to dot 4 subtends at O. [1]

(b) Show that the two lines are equal in length. [1]

9. Chords PQ and RS of a circle with centre O are each 16 cm from O, and $PQ = 24$ cm.

(a) Find RS. Give the reason. [1]

(b) Find the radius of the circle. [1]

10. A chord of a circle is 18 cm long and is 12 cm from the centre. Find the diameter of the circle. Draw a rough diagram that shows the circle, its centre O, the chord AB and the perpendicular OM from O to AB. Mark the given lengths on it. [2]

Section C (3 marks each)

11. AB and CD are parallel chords of a circle with centre O and radius 25 cm. $AB = 40$ cm and $CD = 30$ cm.

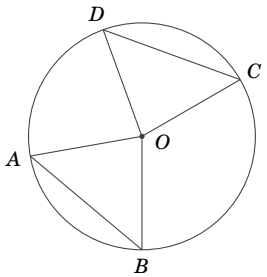
(a) Find the distance of each chord from O. [1]

(b) Find the distance between the chords when they lie on the same side of O. [1]

(c) Find the distance between the chords when they lie on opposite sides of O. [1]

For (b) and (c), draw a rough diagram of each case. Show the circle, its centre O, both chords, and one line through O perpendicular to both chords.

12. AB and CD are equal chords of a circle with centre O. Prove that $\angle OAB = \angle OCD$. [3]



13. Three houses A, B and C stand in a village. $AB = 300$ m, $BC = 400$ m and $\angle ABC = 90°$. The panchayat wants to dig a well at a point W that is at the same distance from all three houses.

(a) Use the scale 1 cm = 50 m. Draw $AB = 6$ cm. At B, draw $\angle ABC = 90°$ and mark C with $BC = 8$ cm. Join AC. [1]

(b) Construct the perpendicular bisectors of AB and BC. Mark their meeting point W. [1]

(c) Measure WA. Find the real distance of the well from each house. [1]

14. A circle has centre O and radius 30 cm. Chord $AB = 48$ cm and chord $CD = 36$ cm.

(a) Without calculating, say which chord is nearer the centre. Give the reason. [1]

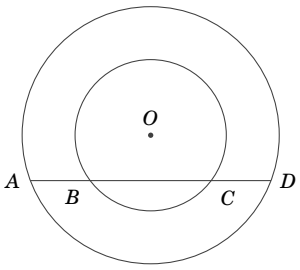
(b) Find the distance of each chord from O, and check your answer to (a). [2]

Section D (5 marks each)

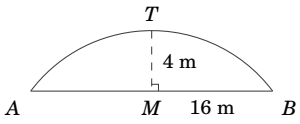
15.

(a) Prove that the perpendicular from the centre of a circle to a chord bisects the chord. [3]

(b) Two circles have the same centre O. A straight line meets the bigger circle at A and D, and the smaller circle at B and C, in the order A, B, C, D (see the figure). Prove that $AB = CD$. [2]



16. The front of a temple gate has an arch. The arch is part of a circle with centre O. The ends A and B of the arch are 16 m apart. The highest point T of the arch is directly above the midpoint M of AB, and $TM = 4$ m.



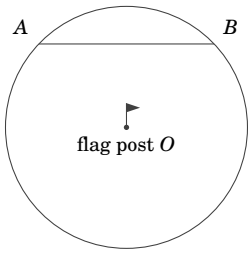
(a) Explain why the centre O lies on the line TM. [1]

(b) Let the radius be r m. Draw a rough diagram that shows the full circle, O, A, B, M and T. Find r. [2]

(c) A straight iron bar CD is fixed across the arch, parallel to AB and 2 m above it, with C and D on the arch. Find the length of the bar. Is your answer in agreement with Theorem 8? Explain. [2]

Section E (case study, 4 marks)

17. Case study. The sports ground of a school is a circle with centre O and radius 29 m. A flag post stands at O. For sports day, the PE teacher draws straight chalk lines from edge to edge of the ground as running tracks. Track AB is 42 m long.



(i) Find the distance of track AB from the flag post. [1]

(ii) Track CD is also 42 m long. Without calculating, write the distance of CD from the flag post. Name the theorem you used. [1]

(iii) Compare $\angle AOB$ and $\angle COD$. Give the reason. [1]

(iv) Track EF is 40 m long. Is EF nearer to the flag post or farther from it than AB? Find its distance from the flag post. [1]